

FPS-3000 SBC — VersaBus address map

Derived from FPS3K_U11_U12_JOIN.bin + emulator measurement. Supersedes the 2026-07 edition; see PDF_ERRATA.md for what changed.

Windows

range	device	contents
\$FF0000-\$FF001F	AP I/F	command reg + bulk data port (see below)
\$FF0040-\$FF00BF	AP I/F	four 32-byte channel windows, \$FF0040+\$20*N
\$FF0200-\$FF021B	XLTR	mode / select / counter / data / status / IRQ mask
\$FF0230-\$FF025F	XLTR	three MC68153-style BIMs, 4 channels each
\$70001C/\$700020	mailbox	host->SBC / SBC->host, one 32-bit word each way
\$400000-\$4FFFFF	chassis	paged memory window; MODE2 = addr bits 20-31

AP I/F command window \$FF0000-\$FF001F

offset	addr	dir	role
+\$00	\$FF0000R/W		command reg; \$8004/\$8005 out. Reads <=0 mean END OF STREAM
+\$04	\$FF0004R		bit 0 = port ready; polled before every transfer (F04B22)
+\$08	\$FF0008R/W		BULK DATA PORT, bidirectional - three modes, below
+\$0E	\$FF000ER		panel command / argument staging

\$FF0008 mode is chosen by the latched opcode \$E5C and bit 5 of the chassis response byte:

\$E5C / bit5	direction	content
\$00 / 0	in	ASCII S-record text, two characters per 16-bit word
\$28 / -	in	binary, word -> staging buffer at \$E58
\$00 / 1	out	binary, staging buffer -> port (this is the WCS upload)

All three use the same handshake: STATUS_IRQ <- \$400, poll bit 15, STATUS_IRQ <- 0.

AP I/F channel windows \$FF0040 + \$20*N [CORRECTED]

offset	ch 1	ch 2	ch 3	ch 4	role
+\$04	\$FF0044	\$FF0064	\$FF0084	\$FF00A4	write port
+\$08	\$FF0048	\$FF0068	\$FF0088	\$FF00A8	32-bit data, HIGH half
+\$0A	\$FF004A	\$FF006A	\$FF008A	\$FF00AA	32-bit data, LOW half
+\$0E	\$FF004E	\$FF006E	\$FF008E	\$FF00AE	command/trigger (\$8000 fires)

+\$08 and +\$0A are one 32-bit register, not data+status: BLK_XFR (F05B0E) reads both each pass; TCBXP1I writes \$0000001B across the pair at F07EC6 then \$8000 to +\$0E. Eight Am29705 16x4 two-port RAMs give exactly 32 bits.

CORRECTION: an earlier edition said \$FF0048 is NEVER READ. It is - by the channel ISR, as \$48(a5) with a5=\$FF0000, so no absolute-address scan sees it. Reads only happen once a channel BIM interrupt is raised, which is why they were missed.

What one channel transaction looks like on a trace [MEASURED]

PC	cycle	meaning
F07EEE	RD +\$0E	read command reg -> RAM \$1066
F07EF6	RD +\$08	read data HIGH -> RAM \$1068
F07EFE	RD +\$0A	read data LOW -> RAM \$106A
F07F10	WR BIM CR \$4F	IRE cleared: mute this channel for the transfer
F07F18	WR +\$08 = \$0000	write data HIGH back
F07F1E	WR +\$0A = d0	write data LOW back
F07F22	WR +\$0E = \$8004	REQUEST-TRANSFER
F07F2E	RD +\$0E (poll)	spin, 1000-iteration timeout in d5

Addresses are XP1I's; the other three channels are byte-identical copies at -\$A00, -\$1400, -\$1E00. XP4I has this ISR too.

The SEPARATE \$8000 path (data pair <- \$0000001B, then \$8000) exists ONLY in XP1I/2/3 and needs command-reg bits 15,14,11 set.

SBC RAM the chassis interacts with

address	owner	role
\$105E	written by CPU	count of channels with a NONZERO command reg; probe at \$F0A202
\$1062 / \$1064	shared	scan state, all four XP tasks
\$1066-\$106A	XP1I	ISR snapshot: {command, data HI, data LO}
\$106C-\$1070	XP2I	same, stride 6
\$1072-\$1076	XP3I	same
\$1078-\$107C	XP4I	same
\$107E / \$1080	shared	scan state
\$10A0-\$10A6	cmd-1 handler	per-channel flag, (ch-1)*2, ch 1-4 ONLY
\$10AA	NOT WRITABLE BY CPU	BI0I1I dispatches on it; needs ch 6, which \$105E forbids
\$10000-\$1FFFF	S-record staging	one 4K x 128-bit WCS bank

XLTR control file \$FF0200-\$FF021B

addr	name	established behaviour
\$FF0200	MODE0	bits 0-7 chassis response byte; b10 ack, b11 valid. Low byte reads 0 when idle
\$FF0202	MODE1	b7 busy (tested 15x), b12 enable (set 8x), b14 control (cleared 12x), b15 arms chassis op
\$FF0204	CHANNEL_SELECT	BIDIRECTIONAL: chassis presents args, SBC returns results from \$E74
\$FF020C	COUNTER	operational value \$4 (7 sites); \$1/\$FF are boot-diagnostic only; readable
\$FF0210	MODE2	chassis page select = address bits 20-31 of the \$400000 window
\$FF0214	DATA_LO	used by phase \$1900 chassis-memory data test
\$FF0216	DATA_HI	mode/page reg, NOT a command reg; gates \$400000 access. Resting value \$C0
\$FF0218	STATUS_IRQ	arm \$400 / poll b15 / clear 0. b4 = BIM POPULATION (16 vs 24 regs)
\$FF021A	IRQ_MASK	reads \$FFF. Bits 5,4,3,2 = XP channels 1,2,3,4 (PanelErrorMaskTable F05C4C)

FPS-3000 SBC — interrupts, protocol, bring-up

Companion to page 1. Every claim here is either read from the ROM or measured in the emulator.

BIM channels \$FF0230-\$FF025F (CR n at +0/+2/+4/+6, VR n at +8/+A/+C/+E)

BIM.ch	CR	value	VR	vector	handler	owner
0.0	\$FF0230	\$5E	\$FF0238	\$41 -> \$104F04930	TCBRDHC	
1.2	\$FF0244	\$5F	\$FF024C	\$45 -> \$114F07EE6	TCBXP1I	
1.3	\$FF0246	\$5F	\$FF024E	\$46 -> \$118F074E6	TCBXP2I	
2.0	\$FF0250	\$5F	\$FF0258	\$47 -> \$11CF06AE6	TCBXP3I	
2.1	\$FF0252	\$5F	\$FF025A	\$48 -> \$120F060CE	TCBXP4I	
2.2	\$FF0254	\$5F	\$FF025C	\$4A -> \$128F05DD6	TCBI01I	

CR bits: 0-2 = IRQ level (0 disables), 4 = IRE enable, 5 = IRAC auto-clear (off), 7 = Flag. \$4F = \$5F with IRE cleared - PanelSendAndWait writes it to suppress a channel's interrupt during a transfer. \$4F has NO connection to \$FF004A.

Chassis -> SBC command protocol (responses arrive on BIM0 ch0)

code	operation	argument (CHANNEL_SELECT)	result
\$0	latch opcode, execute	opcode 0..\$10, or \$28 = bulk	
\$1 / \$41	set address low / high	address half -> \$E58/\$E5A	
\$2 / \$42	set count low / high	count half -> \$E64/\$E66	-
\$3/\$43/\$63	chassis mem write/hi/read	data half -> \$E70/\$E72	\$E70
\$5	arg 0: report AC count	0, or channel number	\$105E
\$7	disable BIM0 ch0 IRQ	-	-
\$9	set data parameter	data half -> \$E68/\$E6A	-
\$B	report \$10010	-	\$E74
\$F	return from interrupt	-	-

Bit 7 of the response byte selects the dispatcher (0 = the 16-entry table at F05102). Bit 6 selects the half for the loaders. Bit 5 selects direction for code \$3 and for the bulk transfer - opposite senses, do not generalise.

Microcode upload — verified end to end in one session

host ASCII S-records -> \$FF0008 -> SRecordDataHandler -> staging \$10010+ -> \$FF0008 -> XLTR -> UNIV FMT -> WCS
S-RECORD ADDRESSES ARE OFFSETS. The handler computes a1 = \$10 + record_address + \$10000 (seed at F051A2, add at F051DC) and range-checks the RESULT to \$10000-\$1FFFF. Usable record range \$0000-\$FFEF -> lands \$10010-\$1FFFF.
There is NO SBC-side WCS bank select: the outbound loop carries only source address and count, no destination.
The chassis is the bus master and places the data; the SBC is a slave conduit and never asserts a transfer request.

Bring-up: the self-test phase beacon

Every power-on self-test writes phase<<8 | subtest to \$FF0204. Scope or read that register during reset and the last value before a hang is the failing subtest - no debugger, no serial port needed.

beacon	suspect	beacon	suspect
\$01xx	CPU flags/arithmetic	\$1000	address-space boundary map
\$02xx	\$1FFF1 b6 <-> \$F70019	\$B1xx-\$14xx	panel-bus IRQs and vectors
\$0300	ROM decode	\$1500	CHANNEL_SELECT read/write
\$0400	RAM address decode	\$1600	XLTR file / BIM population bit
\$0500	(status & \$3F31) == \$3F11	\$17xx-\$18xx	DATA_HI page gating
\$0600	VMOD pattern round-trip	\$1900	chassis memory data lines
\$0700	short-I/O must bus-error	\$1A00	AP I/F window
\$0800	I/O channel	\$2000	RAM address uniqueness
\$0900	MC6840 PTM + \$1FFF1 b7	\$22xx-\$23xx	chassis bulk / A14 decode

Six of Motorola's twelve board-status inputs are USER-DEFINED (Table 1, M68KVM02-3 p.2-59). \$F70019 bits 1-5 are FPS's own wiring and will never appear in a datasheet - infer from firmware, or trace on the board.